MUHAMMED RISWAN P

Junior AI / ML Engineer

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Professional Summary

Data Science professional with 1+ years of hands-on experience building end-to-end machine learning pipelines and MLOps systems. Proficient in Python, Scikit-learn, PyTorch, and the full MLOps stack including Docker, FastAPI, GitHub Actions, DVC, and MLflow. Experienced in computer vision (YOLOv5, U-Net, face recognition), classical supervised and unsupervised ML, and production-grade REST API deployment. Seeking Junior Data Scientist/ML Engineer roles to deliver scalable, data-driven solutions.

Technical Skills

Languages: Python, SQL (PostgreSQL)

Generative AI & LLMs: LangChain, LangGraph, CrewAI, Hugging Face, RAG, Agentic RAG, AI Agents, MCP (Model Context Protocol), A2A Protocol, Vector Search (ChromaDB), Knowledge Graphs, Graph RAG, Supervised Fine-Tuning (SFT, LoRA/QLoRA), Local LLMs (Ollama), Semantic Search, NLP (Classification, Summarization)

Machine Learning: Scikit-learn, Ensemble Methods (XGBoost, Random Forest, Gradient Boosting), Supervised Learning (Regression & Classification), Unsupervised Learning (K-Means, DBSCAN, Hierarchical, PCA), Imbalanced Data (SMOTE), Cross-Validation, Hyperparameter Tuning

Deep Learning: PyTorch, Artificial Neural Networks, CNNs, Transfer Learning (ResNet18, VGG16, MobileNet), Dropout, Batch Normalization, Adam & RMSProp Optimizers

Computer Vision: YOLOv8 (Object Detection), ByteTrack (Multi-Object Tracking), U-Net (Image Segmentation), OpenCV, TorchVision, Face Recognition, Grad-CAM, Inference Optimization

MLOps: Git, DVC, MLflow, Docker, FastAPI, Flask, Pydantic, LLM API Development, GitHub Actions (CI/CD), EvidentlyAI, Cloud Deployment (AWS, GCP, Render)

Analytics & BI: Pandas, NumPy, Matplotlib, Seaborn, Power BI, DAX, Power Query

Statistics: Hypothesis Testing (Z-Test, T-Test, Chi-Square, ANOVA), Bayesian Statistics, Confidence Intervals, Correlation Analysis

Other: BeautifulSoup, Selenium, EDA, Feature Engineering, Data Wrangling

Experience

Data Science & AI Intern Jul 2025 – Present

Bridgeon Solutions, Kozhikode, Kerala

- Built and shipped 10+ end-to-end AI/ML projects spanning classical ML, deep learning, computer vision, MLOps, and Generative AI, applying each domain through production-level implementations.
- Designed and deployed an end-to-end MLOps pipeline using DVC, MLflow, Docker, FastAPI, and GitHub Actions with a live cloud REST API on Render serving real-time ML cluster predictions.
- Implemented automated CI/CD workflows and weekly data drift monitoring using EvidentlyAI with KS-test p-value alerting across 25 features.
- Trained and evaluated 15+ machine learning models across supervised, unsupervised, and deep learning paradigms with MLflow experiment tracking.

Projects

JobFit RAG Assistant GitHub

Python • Hugging Face Transformers • Sentence Transformers • NLP • Semantic Similarity • Prompt Engineering

- Developed an AI-powered assistant to analyze job descriptions against resumes and project portfolios.
- Built a semantic matching engine using Sentence-BERT embeddings and cosine similarity to identify relevant resume skills for job requirements.
- Implemented Hugging Face pipelines for job description summarization and question answering.
- Designed a modular Python package architecture to support future integration of vector search, retrieval, and LLM-powered workflows.

Traffic Flow Analyzer: Vehicle Detection & Traffic Analytics GitHub

Python • YOLOv8 • OpenCV • Pandas • Matplotlib • Docker

- Built an end-to-end computer vision pipeline to detect, track, and count vehicles from traffic surveillance videos using a pretrained YOLOv8n model with ByteTrack multi-object tracking.

- Processed videos frame-by-frame with YOLOv8, producing annotated output with bounding boxes, persistent track IDs, confidence scores, and class labels across 4 vehicle classes (car, truck, bus, motorcycle).
- Implemented virtual line-crossing logic with IN/OUT direction detection using per-track centroid history; resolved 4K video detection failure by tuning inference resolution (`imgsz=1280`).
- Generated structured CSV crossing event logs and a 4-panel matplotlib dashboard covering class breakdown, per-minute flow timeline, traffic mix distribution, and summary statistics.

Customer Segmentation MLOps Pipeline GitHub

Python • KMeans • PCA • FastAPI • Docker • DVC • MLflow • GitHub Actions • EvidentlyAI

- Engineered 3 customer segments from 2,240 records (29 features) using KMeans (k=3) and PCA retaining 90% variance.
- Built a complete MLOps stack with DVC, MLflow, FastAPI, Docker, and GitHub Actions.
- Deployed a live REST API on Render with automated CI/CD workflows and retraining pipelines.
- Implemented dual monitoring using EvidentlyAI reports and KS-test-based p-value alerting across 25 features.

Bank Marketing Subscription Prediction GitHub

Python • Random Forest • Scikit-learn • Flask • Streamlit • Docker • GitHub Actions

- Predicted term deposit subscriptions on 41,188 records with severe class imbalance.
- Evaluated 8 machine learning models and selected a tuned Random Forest achieving ROC-AUC of 0.806.
- Built a complete Scikit-learn Pipeline with preprocessing and leakage prevention.
- Deployed a Flask API, Streamlit dashboard, and Dockerized application with CI/CD automation.

Olist E-Commerce Sales Performance Dashboard GitHub

Power BI • DAX • Power Query • Business Intelligence

- Developed an interactive 4-page Power BI dashboard analyzing 100,000+ orders.
- Created KPI metrics and drill-down reports covering sales performance, delivery operations, customer satisfaction, and seller analytics.
- Identified delivery estimation gaps through root-cause analysis across 100,000+ orders and translated findings into actionable operational recommendations covering logistics, seller performance, and customer satisfaction.

Education

B.Sc. Electronics Graduated: 2025

WMO Arts and Science College, Muttil
University of Calicut